

Auditing AI-Generated Harness Populations: From One-Run Diversity to Stable Complementarity

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Abstract

An automatically generated harness can differ from its peers in source code, execution trace, and correctness outcomes. These differences are often treated as evidence of complementary capability, although each can also arise from an inactive branch, an invalid implementation, or stochastic execution. We study this measurement problem for harnesses wrapping a frozen language model on BIRD text-to-SQL. A discovery population contained 36 outcome-identical pairs among 91 pairs despite source-code differences; trace audits separated absent, untriggered, and broken mechanisms. We then reanalyze a frozen 2×2 study crossing strategy forcing with conformance gating over six builders and three seeds. On 1,169 isolated questions, free ungated generation already had 8.56 ppbare-inclusive oracle headroom. The archived forced-and-gated versus free-and-ungated contrast was -0.16 pp (95% CI $[-0.83, +0.87]$), with no clear improvement. The gate also excludes two allowed prompt-level strategies, so this is a combined screening and conformance contrast rather than a pure verification effect. Repeated executions further show that one-run oracle headroom can be created by the same code. We therefore distinguish source diversity, executed mechanisms, one-run outcome diversity, stable complementarity, and realized selection utility. The current evidence motivates an auditable protocol for measuring these quantities, but does not establish stable additional capability.

1 Introduction

Large language models are increasingly deployed inside harnesses: programs that organize prompts, tool calls, state, verification, and output formatting around a frozen model. Harness-generation systems can produce many candidate programs, creating a natural population-selection problem. A common analysis asks whether the candidates have different source code or whether their correctness vectors differ on a test set. Neither observation answers the practical question that matters for deployment:

Does the population contain additional capability that survives independent executions and can be selected at a fixed cost?

The distinction is easy to lose. A branch may be described but never executed; a repair branch may exist but never be triggered; two valid mechanisms may produce the same answer on the sampled tasks; or one stochastic execution of a single harness may look like several complementary harnesses. In each case, source-level diversity can overstate behavioral diversity, and one-run behavioral diversity can overstate stable complementarity.

We examine these failure modes on BIRD text-to-SQL (Li et al., 2023). The paper has two stages. Phase I is a discovery study that documents the measurement problem in an early generated population. Phase II is a frozen factorial study that asks whether strategy forcing and conformance gating change the observed bare-inclusive oracle headroom under a common generation budget. We report a post-review reanalysis of the archived Phase-II outcomes. This reanalysis corrects the resampling and data-loading chain; it does not add independent executions or repair the original gate.

Our contributions are deliberately narrower than a claim that generated harnesses improve deployment performance:

1. We define separate measurements for source, trace, outcome, and stable complementarity, and give a counterexample showing why oracle headroom is not monotone in population size.
2. We provide a trace-based audit taxonomy and a factorial analysis that makes the implemented admission policy explicit, including its structural exclusion of two allowed prompt strategies.
3. We report the evidence boundary exposed by repeated execution, fixed- K sensitivity, and post-hoc selection analyses. These controls show why a positive one-run headroom value is a necessary diagnostic for routing research, but not evidence of a usable selector.

The central unresolved test is a cost-matched comparison between a population of genuinely different harnesses and a clone population formed by independent executions of the same code. The present data do not contain that control; we keep stable complementarity as an estimand rather than presenting it as a finding.

2 Related Work

Recent work generates, evolves, or routes harnesses around frozen language models, including harness evolution, self-harnessing, just-in-time harness generation, and task-harness co-evolution (Nie et al., 2026; Zhang et al., 2026b;a; Miyai et al., 2026). Concurrent behavior-aware systems use traces or semantic screening inside their search loops (Xu et al., 2026; Chen et al., 2026; Lin et al., 2026; Luo et al., 2026; Wang et al., 2026). Our question is different: what properties does the resulting population have as an outcome matrix, and which generation or admission policy changes those properties?

The distinction matters because a behavior-aware gate inside an evolution loop does not by itself establish population complementarity. Likewise, portfolio routing and algorithm-selection work assumes that candidate systems provide reliable conditional value (Rice, 1976; Xu et al., 2008; Kotthoff, 2014); our study measures the conditions that a future router would need to inherit. HELIX and related work emphasize sibling-harness outcome matrices and complementarity (Fan & Huang, 2026), while studies of harness sensitivity show that no single harness need dominate across instances (Gupta et al., 2026; Cho, 2026). We build on these ideas by separating one-run coverage from repeat-stable value and by auditing the instrument that decides whether a mechanism was actually present.

3 Measurement Framework

Let H wrap a frozen target model M , receive an item x , make zero or more model calls and tool executions, and return an answer scored by a judge. Let H_0 be the bare single-call baseline and $Y(x, H, M) \in \{0, 1\}$ its correctness. The item-level marginal value is

$$\Delta(x, H, M) = Y(x, H, M) - Y(x, H_0, M).$$

For a population P , we report three one-run quantities. Pairwise outcome disagreement is the fraction of items on which two members have different correctness. Union repair is the fraction of bare errors repaired by at least one member. Bare-inclusive oracle headroom is

$$H(P) = \text{Acc} \left(x \mapsto \max_{H \in P \cup \{H_0\}} Y(x, H, M) \right) - \max_{H \in P \cup \{H_0\}} \text{Acc}(H).$$

Positive $H(P)$ means that, in this particular outcome matrix, different members cover different items beyond the best fixed member. It does not imply that a selector can identify those members before seeing the answer.

Headroom is not monotone in $|P|$. If two members solve complementary halves, oracle accuracy can be 100% and best-fixed accuracy 50%. Adding a member that solves every item

leaves oracle accuracy at 100% but raises best-fixed accuracy to 100%, reducing headroom to zero. Thus fixed- K analyses are sensitivity analyses, not proofs of a population-size mechanism.

For independent executions, let $p_{h,x}$ be the probability that harness h solves item x . A repeat-stable analogue is

$$H_{\text{stable}} = \frac{1}{n} \sum_x \max_h p_{h,x} - \max_h \frac{1}{n} \sum_x p_{h,x}.$$

A clone population whose members are the same code has identical $p_{h,x}$ and therefore zero stable headroom, even though separate one-run executions can produce positive oracle headroom. Estimating this quantity requires independent repeats and a selector evaluation that is separated from the observations used to fit it.

4 Phase I: Discovery Audit

Phase I uses GLM-5.3-Flash at temperature zero on 60 BIRD questions. It is discovery evidence: 24 items overlapped protocol development, the human positive control was designed after trace inspection, and the legacy judge was a conservative proxy. These conditions support diagnosis, not confirmation.

The day-1 population contains bare, react, and 12 generated candidates. Across all 14 harnesses, 36 of 91 pairs have identical correctness vectors and mean pairwise disagreement is 3.3%. Among the 12 generated candidates alone, 21 of 66 pairs are identical and mean disagreement is 3.8%. The candidate pair code-similarity mean is 0.143 under the original normalized-character metric. In the full population, the token-sequence metric used in Figure 1 has mean 0.190 and Spearman $\rho = -0.010$ with outcome disagreement. Pairs share harnesses, so this is a descriptive summary rather than an independent sample test.

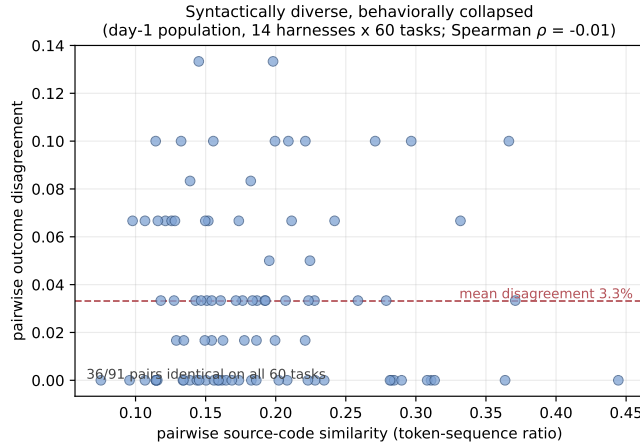


Figure 1: Phase-I discovery audit. Each point is a pair among 14 harnesses on 60 tasks. Source-code similarity and outcome disagreement are distinct measurements; $\rho = -0.01$ is descriptive, and 36/91 pairs are outcome identical.

Trace inspection identifies three actionable classes. T1 mechanisms are described but absent from control flow: six of 12 candidates make one model call and execute no artifact. T2 mechanisms are present but untriggered: in all 10 audited react tasks, the first SQL succeeds, so conditional repair does not run. T3 is a broken multi-turn path: one candidate returns prose as final SQL and turns some bare-correct items into errors. These classes require different controls. A mechanism gate must test execution, a repair claim must be tested with controlled faults, and a multi-turn harness must validate its executable output.

Hand-written positive controls show that task-side headroom exists in the observed matrix: forcibly distinct execution paths give 6.67 ppon 60 items and 7.28 ppon a 151-item probe. Because the control was designed after inspection, these values establish available headroom, not an unbiased effect estimate.

5 Phase II: Frozen Factorial Study

Phase II crosses two factors over six builders and three seeds. The free arm allows the builder to invent a mechanism; the forced arm assigns one of eight named strategies. The ungated arm applies neutral validity checks; the gated arm additionally requires the frozen conformance suite. We call the four arms II-A (free/ungated), II-B (free/gated), II-C (forced/ungated), and II-D (forced/gated). An open-mechanism arm II-E is secondary and outside the factorial.

The conformance suite drives scripted scenarios and checks recorded traces, including paired counterfactuals whose correct answers differ. It was calibrated before Phase-II generation with three positive controls and two negative impostors. The implemented gate maps both `hint_guard` and `format_guard` to a no-mechanism result. Therefore II-D versus II-A combines strategy forcing, structural screening, and conformance verification; it cannot identify verification alone.

Each slot permits at most three raw builder attempts and stops at the first accepted candidate. Realized attempts were 269, 255, 273, and 324 for II-A through II-D against a 432-attempt arm cap; realized token cost is not fully recovered. Slots are not topped up after failure, so generation reliability is part of the policy and is reported separately from population quality.

The target is GLM-5.3-Flash. The primary set has 1,169 questions from nine databases never opened during development, with a disclosed 18-item smoke-test exception. All 18 builder-seed cells enter the primary analysis, including empty populations. The official BIRD execution scorer is primary; the legacy judge is a conservative secondary check. A 400-item core, frozen before outcomes, supports the four-arm secondary analysis.

Phase-II confirmatory design: database-level isolation, 2x2 factorial + open-mechanism arm, three-layer admission

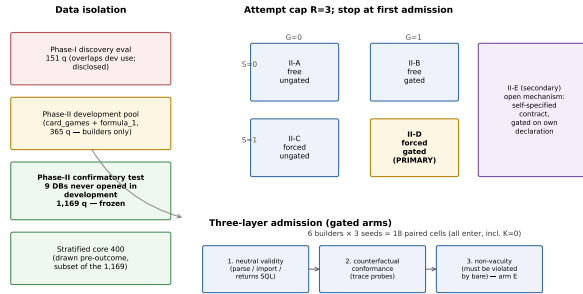


Figure 2: Frozen Phase-II design. Strategy forcing and conformance gating are crossed under a common raw-attempt cap; realized costs and admitted population sizes can differ. The gate is a trace instrument with neutral validity, counterfactual conformance, and non-vacuity checks.

The primary estimand is the paired cell difference $H(\text{II-D}) - H(\text{II-A})$ on the full set. We use shared database and within-database task resampling, generation-seed resampling within builder, and a sign-flip reference test conditional on observed tasks. The factorial secondary estimands on the common core are the gate main effect, strategy main effect, and their interaction. These are conditional analyses of archived executions, not randomized tests of an abstract gate.

6 Results

6.1 Primary contrast

Table 1 reports the corrected reanalysis. Free ungated generation already produces 76.14% oracle accuracy, 67.58% best-fixed accuracy, and 8.56 ppheadroom. Forced gated generation produces 75.70% oracle accuracy, 67.30% best-fixed accuracy, and 8.40 ppheadroom. The paired headroom contrast is -0.16 pp with 95% CI $[-0.83, +0.87]$ and one-sided sign-flip $p = 0.647$; cell contrasts range from -4.36 to $+2.74$ pp.

Metric	II-A	II-D
Candidate count	5.78	3.83
Oracle accuracy (%)	76.14	75.70
Best-fixed accuracy (%)	67.58	67.30
Headroom (pp)	8.56	8.40
D-A headroom (pp)	-0.16	
Unadjusted bootstrap 95% CI	[-0.83, +0.87]	
One-sided sign-flip p	0.647	

Table 1: Corrected primary analysis of the archived intervention. Bare is retained in oracle and best-fixed calculations. The interval reflects the versioned task and generation-seed resampling chain, conditional on the six builders and archived execution records.

The interval does not establish equivalence, but it provides no clear evidence that the implemented forced-and-gated policy improves headroom. Because the gate excludes allowed prompt transformations, the result should not be described as a pure conformance effect.

6.2 Factorial and fixed-population sensitivity

On the common 400-item core, the corrected gate main effect is -0.69 pp with Holm-adjusted $p = 0.079$; the strategy effect is $+0.42$ pp (adjusted $p = 0.428$) and the interaction is -0.87 pp (adjusted $p = 0.428$). The fixed- K gate sensitivity is -0.31 pp with 95% CI $[-1.07, +0.40]$. The reduction in a subsampled contrast is a sensitivity result; it is not a causal decomposition of how much population size mediated the original contrast.

Contrast	Effect (pp)	95% CI (pp)	Raw p	Holm p
Gate	-0.69	$[-1.56, +0.20]$	0.026	0.079
Strategy	+0.42	$[-0.32, +1.51]$	0.263	0.428
Interaction	-0.87	$[-2.83, +0.65]$	0.214	0.428

Table 2: Corrected factorial reanalysis on the common core. Gate, strategy, and interaction estimands are defined by the four arm means. Raw tests use cell sign flips and Holm correction covers the three secondary contrasts.

For the full primary set, oracle accuracy changes by -0.44 pp and best-fixed accuracy by -0.28 pp from II-A to II-D. The difference in these components is reported explicitly because headroom is an arithmetic diagnostic, not a direct measurement of realized routing utility. Candidate counts also differ: K_{cand} excludes bare and $K_{\text{total}} = K_{\text{cand}} + 1$.

Arm	K_{cand}	K_{total}	Bare (%)	Best (%)	Oracle (%)	H (pp)
A	5.78	6.78	63.25	66.00	74.51	8.51
B	6.11	7.11	63.25	66.12	74.39	8.26
C	5.61	6.61	63.25	65.64	75.01	9.38
D	3.83	4.83	63.25	65.65	73.90	8.25

Table 3: Same-core decomposition. All arms share the same bare outcomes. Headroom is oracle accuracy minus best-fixed accuracy; it is not selector accuracy.

6.3 Execution variability and selection boundary

Across three cache-off repeats, 14.5% of bare tasks changed correctness at least once. A canonical R2 replay found 2,391 same-harness verdict flips among 28,000 comparisons (8.54%). The cache-off subset contrast changed from $+0.42\text{ pp}$ to $+0.81\text{ pp}$; its change was $+0.39\text{ pp}$ with 95% CI $[-0.95, +1.63]$. These are instability measurements, not standard errors or a noise floor for the primary effect. They do not show that every contrast is unchanged.

As a direct diagnostic, three executions of the identical bare source treated as if they were three population members yield 72.25% one-run oracle accuracy and 65.75% best-fixed accuracy, a 6.50 ppartificial headroom gap on the same 400 tasks. The source hash is identical, so this result demonstrates only that execution sampling can create one-run headroom.

The Phase-I leave-one-harness-out predictor did not beat the best fixed member in the tested DeepSeek population (-3.4 pp); the Qwen estimate was $+0.9\text{ pp}$ with CI $[-5.2, +6.9]$. Database-stratified task-held-out controls reduced repair AUROC to 0.512. A post-hoc diversity-selection analysis at $k = 8$ changes headroom by $+0.78\text{ pp}$, with oracle accuracy changing by $+0.35\text{ pp}$ and best-fixed accuracy by -0.42 pp . These analyses are descriptive and single-execution; they do not establish a deployable router or practical selection gain.

7 Discussion

The results support a measurement conclusion. Code differences, executed mechanisms, one-run correctness differences, repeat-stable complementarity, and realized selector value are different objects. The Phase-I population makes the first boundary visible. The Phase-II archive tests one policy contrast but does not remove the second boundary: observed oracle headroom can contain execution randomness, and the gate’s structural exclusions prevent a pure verification interpretation.

This framing also changes how population size should be interpreted. Increasing the number of candidates can increase oracle coverage, leave it unchanged, or reduce headroom if a strong fixed member improves the best-fixed comparator. Similarly, a K-matched difference tells us how the estimand changes after subsampling; it does not identify a causal share attributable to population size.

The most informative next experiment is a cost-matched clone control. It should use the same raw pool, interleave real and clone executions, run each code path under cache-on and cache-off conditions, and reserve independent repeats for validation. It should report absolute oracle accuracy, best-fixed accuracy, realized selector accuracy if a selector is specified, candidate count, and cost. A gate calibration set should separately include legal single-call rewrites, conditional repair, unconditional extra calls, discarded feedback, position-based voting, and untriggered repair. This is necessary to distinguish an absent mechanism, an untriggered mechanism, a nonresponsive mechanism, and an implementation failure.

The evidence is limited to one main target, six fixed builders, and three generation seeds. Phase I contains development overlap and a post-trace human control. The corrected analysis is post-review and does not add executions. The one-seed MATH-500 probe in the supplement extends the outcome comparison to a second domain, but has no clone control, repaired-gate factorial, or independent stable-complementarity test. Partial provider

acquisitions are retained for provenance and missingness accounting and are excluded from population-performance claims.

8 Conclusion

AI-generated harness populations should be audited at the level of what they execute and what remains reliable across repeats. In our archived BIRD study, the implemented forced-and-gated policy does not show a clear headroom gain, and the available one-run headroom is not sufficient to claim stable complementarity. The practical contribution is an evidence boundary: a population becomes a candidate for reliable routing only after it passes mechanism calibration, cost matching, repeat validation, and independent selection evaluation.

AI use statement

This work studies AI-generated artifacts and also used generative AI tools for experimental code, model-call workflows, data analysis, figures, and manuscript drafting. The archived tables and result values in this rewrite are read from the versioned reanalysis artifacts; historical exploratory results retain their stated provenance. AI tools also assisted with research design and revision. The authors are responsible for the final scientific decisions, claims, and submission.

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A Reproducibility Notes

All main tables in this rewrite use the shared versioned macros and table artifacts in the paper directory. The archived analysis distinguishes source hash, task, repeat, cache condition, arm, membership, source file, and conflict policy. Reproduction should begin from the canonical input order and should report missing, duplicate, and conflicting records before computing aggregate metrics. The full protocol changelog, gate controls, extraction audit, cross-domain probe, and W1 sensitivity analysis remain in the existing supplementary source files and must be synchronized before submission.